PART A: SHORT QUESTIONS **[10 Marks]**

Answer the following 5 statements by first stating whether it is correct or incorrect and then followed by a ONE-sentence motivation (reason) of the answer.

1. The advantage of low-fidelity prototyping addresses screen layout issues while its disadvantage addresses inefficient for proof-of-concept designs.
2. Evaluation methods which include asking users and experts must be performed using both control and natural settings situation.
3. Usability testing may involve user satisfaction questionnaires and interviews to provide data about users' opinions and also compare of products or prototypes.
4. Fitts' Law is one of the most widely used techniques to evaluate system for a handheld and mobile devices which the time to locate an object is important.
5. The data analysis that can be done depends on the approach of the performed data gathering. Either qualitative or quantitative data may be gathered from any of the two data gathering approaches, including questionnaire and interview.

PART B: STRUCTURED QUESTIONS [33 Marks]

This part consists of **FOUR** short and structured questions. The marks for each part of the question are as indicated.

QUESTION 1 (11 marks)

You are asked to take the responsibility of designing a machine. This machine can be worked as a printer and a scanner. It can also be used as a fax and a copy machine. It can be installed at home, or in small offices.

a) To ensure the designing of the product fulfils the usability criteria, you decide to include the four basic activities in Interaction Design in your project plan. What are these **FOUR** basic activities? List all of them. (4 marks)

b) One of the practical issues in a project is to identify who are the users/stakeholders of a product. In the case of your project stated above, who would be your users/stakeholders? List **FOUR** of them. (4 marks)

c) **List and briefly** explain **THREE** quality thresholds that will help you choose among the design alternatives for your product. (3 marks)

QUESTION 2 (6 marks)

Low fidelity prototyping (LFP) uses medium which is unlike the final medium. It is often used with scenarios, bringing more detail, and a chance to role play. One of the approaches used for LFP is the *Wizard of Oz*.

a) Describe what is the *Wizard of Oz* (4 marks)

b) Give an example to how and when this approach is best used. (2 marks)

QUESTION 3 **(4 marks)**

Gestalt principles or laws are rules that describe how the human eye perceives visual elements. These principles aim to show how complex scenes can be reduced to more simple shapes. They also aim to explain how the eyes perceive the shapes as a single, united form rather than the separate simpler elements involved. Given the following figures A.3a, A.3b, A.3c and A.3d, name the correct Gestalt principle used, respectively.

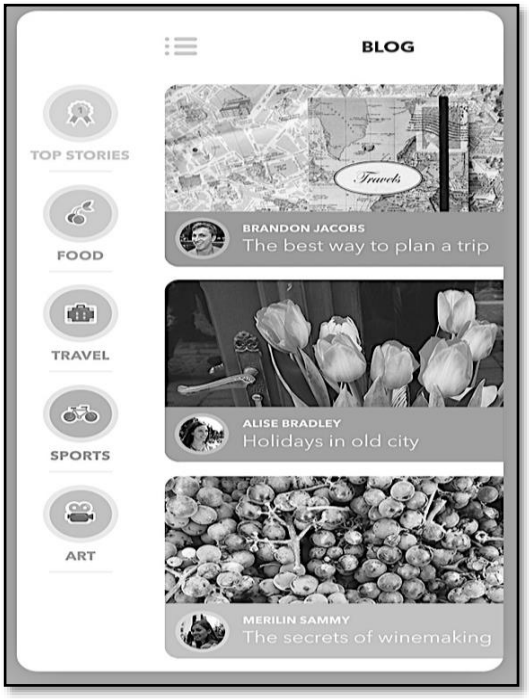


Figure A.3a.



Figure A.3b.

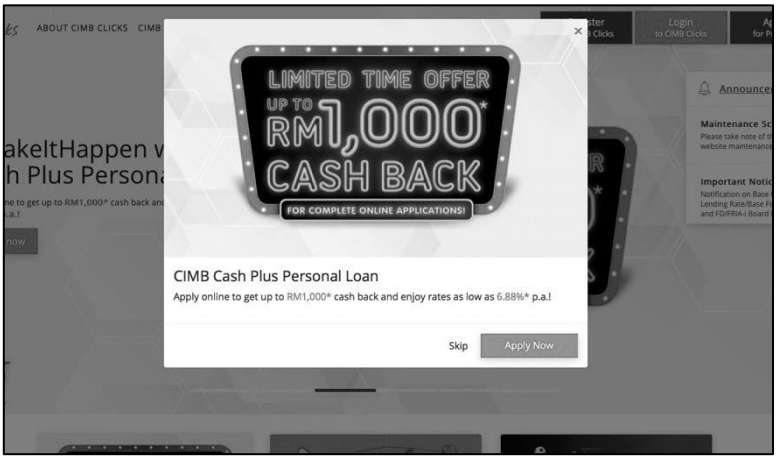


Figure A.3c.

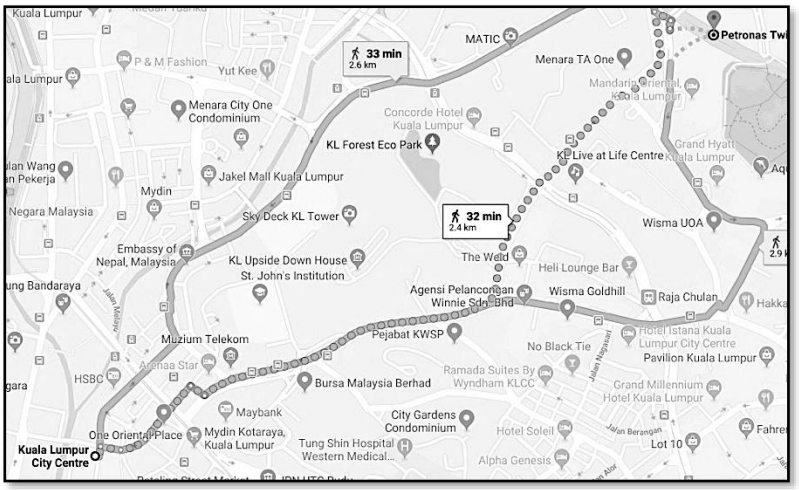


Figure A.3d.

QUESTION 4 **(12 marks)**

Heuristic evaluation and usability testing are two different techniques for finding usability problems.

a) Explain the difference between Heuristic Evaluation and Usability Testing. (4 marks)

b) List and describe **TWO** of Nielsen's heuristic evaluation (design guideline principle) by providing **ONE** example to support each answer. (8 marks)

PART B: LONG QUESTION **[67 Marks]**

This part consists of **TWO** long questions. The mark for each part of the question is as indicated.

QUESTION 1 **(34 marks)**

You are currently working with a company that provides digital solutions. The past products involve enterprise systems, and mobile applications. Towards the 4th Industrial Revolution (4IR), the company is now prepared to embrace and take a leap in delivering the Internet of Thing (IoT) digital services.

Based on IoT Agenda[1], the Internet of Things (IoT) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. See Figure B.1.

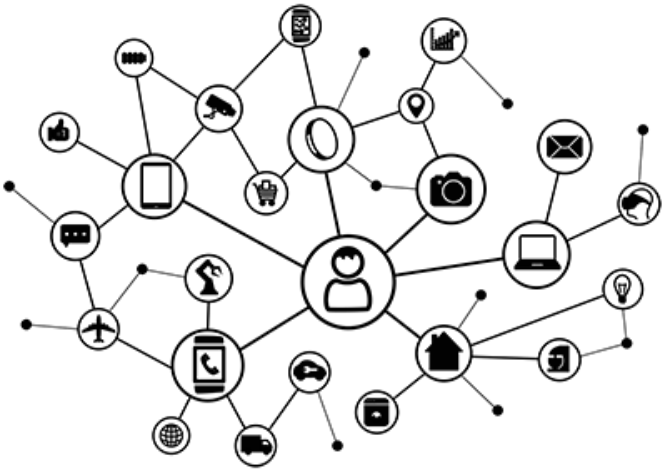


Figure B.1. Internet of Things (credit Champion Solutions Group)

The company understands that the rush decision to develop any IoT project is not wise. Hence, your team is responsible to perform an insight study to understand about the users, their behaviours and their needs, before arriving to recommendations. Your first IoT project will be implemented in the context of higher institutions.

Based on the above, answer the following questions.

[1] "What is Internet of Things (IoT)? – Definition from WhatIs.com." http://internetofthingsagenda.techtarget.com/definition/Internet-of-Things-IoT. Accessed 26 Nov. 2019.

a) What is the importance of understanding your users before embarking with IoT implementation? (2 marks)

b) You only have a week to perform the insight study. What TWO techniques would you find to be the most suitable to gather information about your users? Why? Elaborate by including the recorded data type and equipment used. (7 marks)

c) How do you represent your users from the data gathered? What are the most important elements or characteristics to capture? (3 marks)

d) Based on the context of IoT at higher institution, distinguish TWO targeted users you found to be most important to be considered in (d), and suggest TWO goals. (5 marks)

e) From your answers in (d), provide TWO scenarios by sketching via storyboarding, how each user achieving their goals. Storyboard should convey setting (people, environment, task), sequence (steps, shows interactions), and satisfaction. (12 marks)

f) For each scenario in (e), justify the reasons why you pursue for the chosen interaction and interface types. You may choose 1 interface and 1 interaction type for each scenario. (5 marks)

QUESTION 2 **(33 marks)**

Figure B.2 shows the front page of the Universiti Teknologi Malaysia' Counselling Center website. You have been asked to re-design or design a new version of the page that can be viewed correctly on multiple screens including desktop and mobile phone.

a) How does the page measure up against UI/UX design principles? Which part of it is good? Which part of it is bad? Apply any **THREE** design principles/heuristics that you have learned to support your answer. (6 marks)

b) You have decided to setup a survey asking prospective student, current student, staff, alumni and visitor on how they use the website to identify the most commonly used elements. How would you encourage good questionnaire response? List **FOUR** ways to do so. (4 marks)

c) Describe **THREE** specific problems or research designs issues that need to considered when designing for mobile phone screen. (6 marks)

d) How would you evaluate your new design? Give a **thorough** description on how you would execute a usability study for empirical evaluation. Your answer should also include specification on what to measure including independent and dependent variables, and under what conditions, and how to avoid order effects. (17 marks)

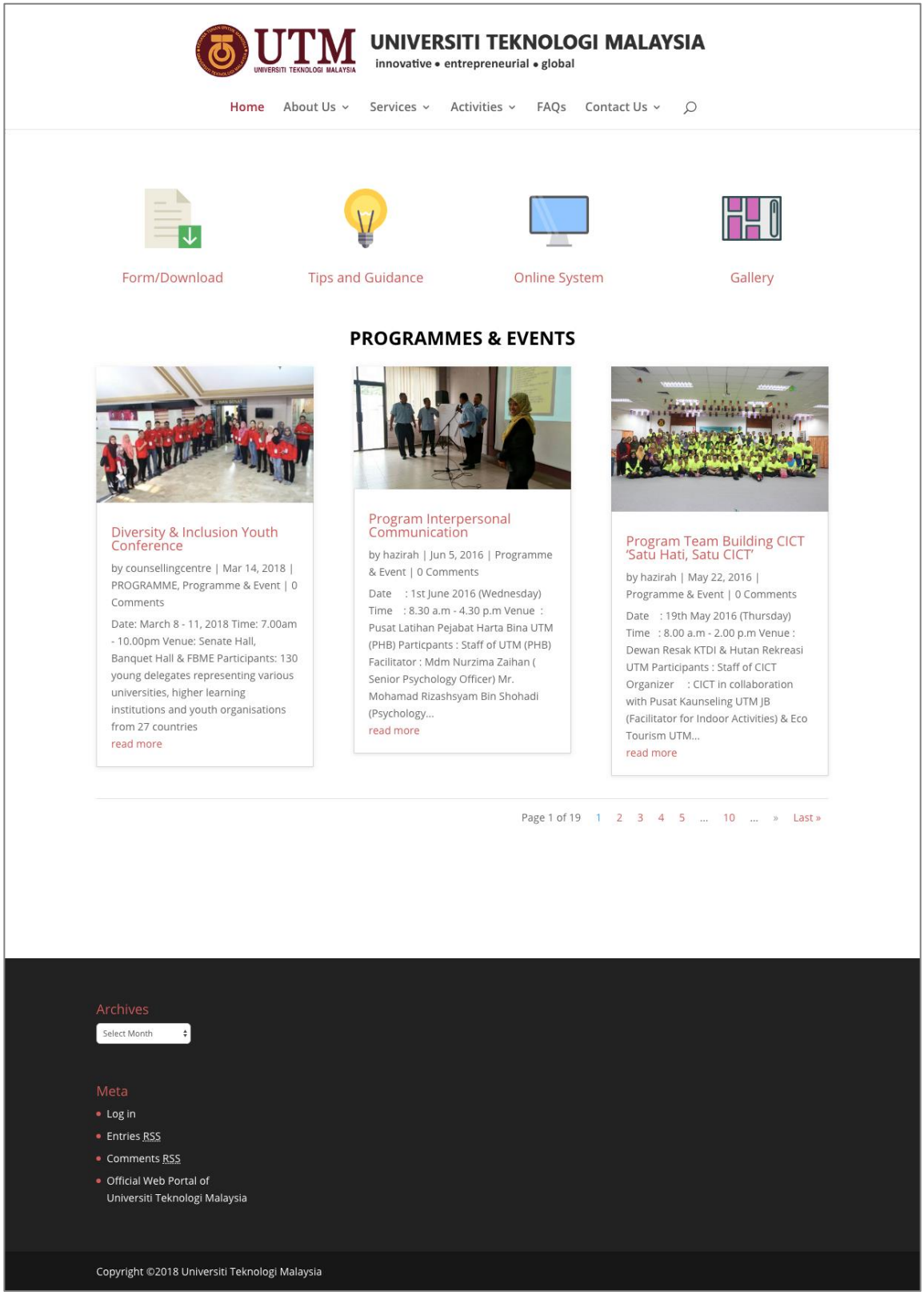


Figure B.2: The front page of Universiti Teknologi Malaysia's Counselling Center website

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